



BAULKHAM HILLS HIGH SCHOOL

2017 HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A reference sheet is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for careless or badly arranged work

Total marks – 100

Exam consists of 10 pages.

This paper consists of TWO sections.

Section I – Page 2-4 (10 marks)

- Attempt Question 1-10
- Allow about **15 minutes** for this section

Section II – Pages 4-10 (90 marks)

- Attempt questions 11-16
- Allow about **2 hours and 45 minutes** for this section

Section I

10 marks

Attempt questions 1-10

Allow about 15 minutes for this section.

Use the multiple choice answer page for questions 1-10

1. Simplify $\frac{x^2-4}{x^3-8}$

- (A) $\frac{x-2}{x^2-2x+4}$
(B) $\frac{1}{x+2}$
(C) $\frac{x+2}{x^2+4x+4}$
(D) $\frac{x+2}{x^2+2x+4}$

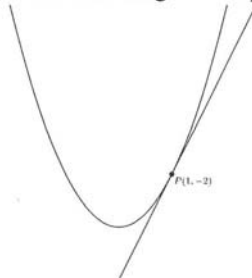
2. A yacht sailed directly from A to B on a bearing of 196°T . To sail from B back to A, the bearing should be:

- (A) 016°T
(B) 074°T
(C) 164°T
(D) 196°T

3. Which of the following defines the domain of the function $f(x) = \frac{1}{\sqrt{4-x^2}}$?

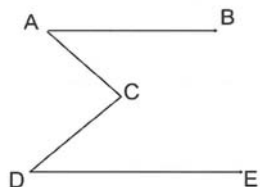
- (A) $x < -2$ or $x > 2$
(B) $x < -2$
(C) $x > 2$
(D) $-2 < x < 2$

4. The diagram shows a concave up parabola with a tangent at the point $P(1, -2)$.



Which of the following could be the equation of the normal at P?

- (A) $x - 3y + 5 = 0$
 (B) $2x - 3y + 1 = 0$
 (C) $x + 3y + 5 = 0$
 (D) $x + 3y - 5 = 0$
5. In the diagram, AB and DE are parallel, $\angle BAC = 30^\circ$ and $\angle CDE = 55^\circ$.



What is the size of $\angle ACD$?

- (A) 45°
 (B) 65°
 (C) 85°
 (D) 95°
6. What is the derivative of $\frac{e^x}{e^x + 1}$?

- (A) $\ln(e^x + 1)$
 (B) $2\ln(e^x + 1)$
 (C) $\frac{2e^{2x} + e^x}{(e^x + 1)^2}$
 (D) $\frac{e^x}{(e^x + 1)^2}$

7. The quadratic equation $3x^2 - 5x + 1 = 0$ has roots α and β . Which of the following statements is true?

- (A) $2\alpha\beta = -\frac{4}{3}$
 (B) $\alpha^2 + \beta^2 = \frac{13}{9}$
 (C) $2\alpha + 2\beta = \frac{10}{3}$
 (D) $\alpha^2\beta^2 = \frac{2}{9}$

8. In $\triangle ABC$, $AB=5\text{m}$, $AC=8\text{m}$ and $\angle ACB = 40^\circ$. If $BC=d$ metres, which equation could be solved to find the length of BC ?

- (A) $d^2 = 5^2 + 8^2 - 2 \times 5 \times 8 \sin 40^\circ$
 (B) $d^2 = 5^2 + 8^2 - 2 \times 5 \times 8 \cos 40^\circ$
 (C) $5^2 = 8^2 + d^2 - 2d \times 8 \sin 40^\circ$
 (D) $5^2 = 8^2 + d^2 - 2d \times 8 \cos 40^\circ$

9. The values of x for which the geometric series $2 + 4x + 8x^2 + \dots$ has a limiting sum are:

- (A) $x < \frac{1}{2}$
 (B) $x \leq \frac{1}{2}$
 (C) $-\frac{1}{2} < x < \frac{1}{2}$
 (D) $-\frac{1}{2} \leq x \leq \frac{1}{2}$

10. What is the number of solutions for the equation $x = 10 \sin x$?

- (A) 4
 (B) 5
 (C) 6
 (D) 7

END OF SECTION I

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question on the appropriate page in the writing booklet.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start on the appropriate page in the answer booklet

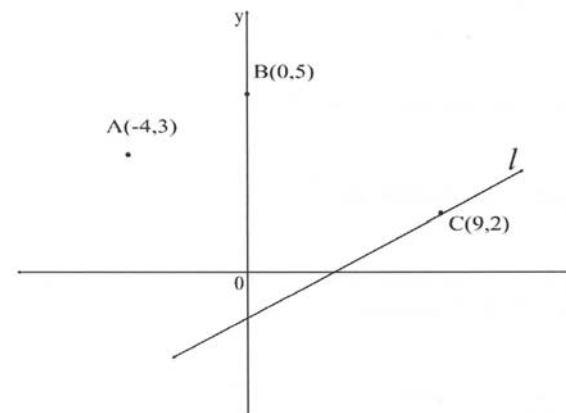
- (a) Factorise $2x^2 + x - 28$ 2
- (b) Solve $|x - 1| = 2x - 3$ 2
- (c) Find $\int \sqrt{3x - 4} \, dx$ 2
- (d) If $\tan \theta = \frac{5}{8}$ and $\sin \theta < 0$ find the exact value of $\operatorname{cosec} \theta$. 2
- (e) The points $A(4, 12)$, $B(8, 6)$ and $C(-4, k)$ are collinear. Find the value of k . 2
- (f) The quadratic equation $2x^2 - (4p + 1)x + 2p^2 - 1 = 0$ has two roots which are equal in magnitude but opposite in sign. Find the value of p . 2
- (g) Find the equation of the curve $y = f(x)$ given that $y = 5x - 7$ is a tangent to this curve, and its gradient function is $f'(x) = 4x - 3$. 3

Question 12 (15 marks) Start on the appropriate page in the answer booklet

(a) Differentiate with respect to x :

- (i) $(x^2 + 4)^5$ 2
- (ii) $e^x \log_e x$ 2
- (iii) $\frac{\sin x}{x-1}$ 2

(b) The diagram shows the points $A(-4, 3)$, $B(0, 5)$ and $C(9, 2)$, and the line l which passes through C and is parallel to AB .



Copy the diagram into your answer booklet.

- (i) Find the length AB . 1
- (ii) Show that the equation of line l is $x - 2y - 5 = 0$ 2
- (iii) Find the coordinates of point D , where line l meets the x -axis. 1
- (iv) Prove that $ABCD$ is a parallelogram. 2
- (v) Find the perpendicular distance from point B to line l . 2
- (vi) Hence or otherwise, find the area of parallelogram $ABCD$ 1

Question 13 (15 marks) Start on the appropriate page in the answer booklet

(a) Evaluate $\int_1^3 \frac{5x}{x^2 + 1} dx$

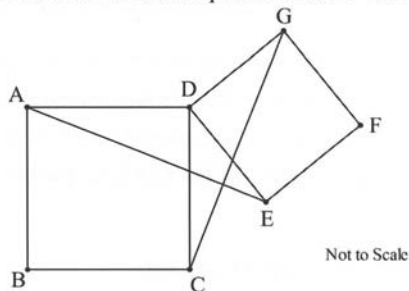
2

(b) For which values of k does the equation $3x^2 + kx + 5 = 0$ have no real solutions?

3

(c) In the diagram below, $ABCD$ and $DEFG$ are two squares. Prove $AE = CG$.

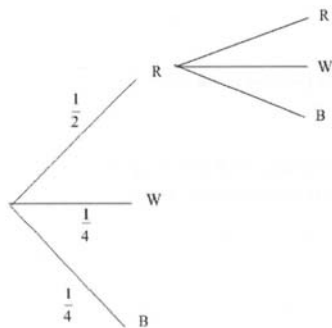
3



(d) A jar contains 2 red, 1 black and 1 white marble only. Ken randomly selects two marbles in succession (without replacement), and places them in a bag.

(i) Copy and complete the probability tree to show all possible outcomes.

2



(ii) Find the probability that both of the selected marbles are red.

1

(ii) If one of the marbles falls out of the bag and we see that it is white, what is the probability that the marble still in the bag is red?

1

(c) Evaluate $\sum_{n=1}^{20} (3^n + 2n - 1)$

3

Question 14 (15 marks) Start on the appropriate page in the answer booklet

(a) Given $\log_3 2 = p$ and $\log_3 5 = q$, find in terms of p and q :

(i) $\log_3 12.5$

1

(ii) $\log_5 2$

1

(b) The velocity of a particle is given by $v = 1 - 2 \cos t$ for $0 \leq t \leq 2\pi$, where v is measured in metres per second and t is measured in seconds.

(i) At what times during this period is the particle at rest?

2

(ii) What is the maximum velocity of the particle during this period?

2

(iii) Sketch the graph of v as a function of t for $0 \leq t \leq 2\pi$

2

(iv) Calculate the total distance travelled by the particle between $t = 0$ and $t = \pi$

3

(c) For the parabola $x^2 - 10x - 16y - 7 = 0$, find :

(i) the coordinates of the vertex.

2

(ii) the equation of the directrix.

1

(iii) the coordinates of the focus.

1

Question 15 (15 marks) Start on the appropriate page in the answer booklet

(a) Radium is a radioactive substance which decays over time, with a half-life of 1600 years. (Note: the half-life of a substance is the time taken for it to decay to half its mass).

The mass M remaining at time t years is given by:

$$M = M_0 e^{-kt}$$

where M_0 and k are constants.

(i) Evaluate k , expressing your answer correct to 3 significant figures.

2

(ii) If a piece of radium in a laboratory has a mass of 200 grams, what would have been its mass 1000 years ago? Answer correct to the nearest gram.

2

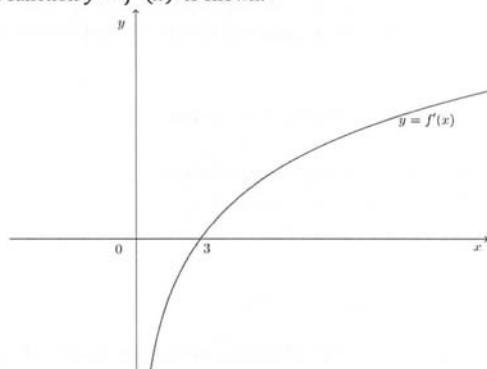
(b) Prove that $(1 - \cos \theta)(1 + \sec \theta) = \sin \theta \tan \theta$

2

Question 15 continues on the next page.

Question 15 (Continued)

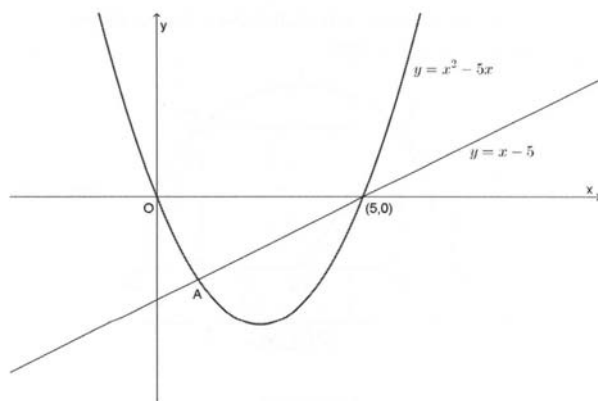
- (c) A sketch of a gradient function $y = f'(x)$ is shown.



Sketch the curve $y = f(x)$, given $f(3) = 0$.

3

- (d) The graphs of $y = x - 5$ and $y = x^2 - 5x$ intersect at the points (5,0) and A, as shown in the diagram.



- (i) Show that A has coordinates (1,-4).
 (ii) Find the area between the line $y = x - 5$ and the curve $y = x^2 - 5x$.
 (e) A sector of a circle is enclosed by two radii of length 10cm and an arc of length 14cm. Find the area of the sector.

1

3

2

Question 16 (15 marks) Start on the appropriate page in the answer booklet

- (a) Karina borrows \$80 000 to start a business. She plans to repay the loan in equal monthly instalments of \$M at 0.5% per month reducible interest. Interest is calculated and charged just before each repayment.

Let A_n = amount owing after n repayments (in dollars)

- (i) Show that the amount owing after 3 repayments is
 $A_3 = 80000(1.005)^3 - M(1.005^2 + 1.005 + 1)$

2

- (ii) Show that

$$A_n = 1.005^n(80000 - 200M) + 200M$$

2

- (iii) If she wishes to repay the loan in 10 years, calculate the amount of each instalment.

2

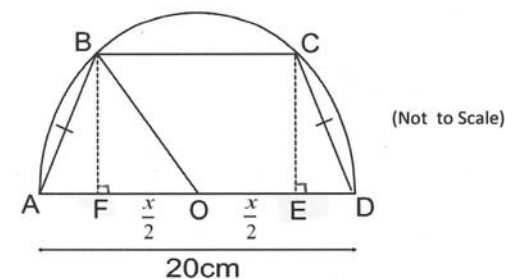
- (iv) If she can only repay \$700 per month, how long will it take her to repay the loan? Answer correct to the nearest year.

2

- (b) Solve $\sin(x + \frac{\pi}{6}) = \frac{\sqrt{3}}{2}$ for $0 \leq x \leq 2\pi$

2

- (c) ABCD is a trapezium inscribed in a semicircle of diameter 20cm, as shown. $AB = CD$ and O is the centre of the semicircle.



- (i) If $EO = OF = \frac{x}{2}$, show that $BF = \frac{1}{2}\sqrt{400 - x^2}$
 (ii) Show that the area $A \text{ cm}^2$ of the trapezium ABCD is given by $A = \frac{1}{4}(x + 20)\sqrt{400 - x^2}$
 (iii) Hence find the length of BC such that the area of the trapezium ABCD is a maximum

1

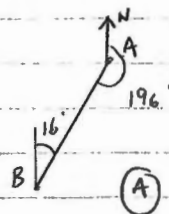
1

3

End of exam

1. $\frac{(x-2)(x+2)}{(x-2)(x^2+2x+4)}$

(D)



(A)

3. $4 - x^2 > 0$

$4 > x^2$

$x^2 < 4$

$-2 < x < 2$ (D)

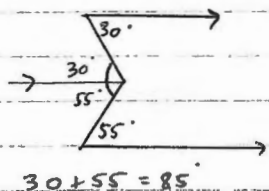
4. $m_{\text{tang.}} > 0$ $m_{\text{Norm.}} < 0$

C or D

but must pass through (4, 2)

Test C: $1 - 6 + 5 = 0$

Test D: $1 - 6 - 5 \neq 0$ (C)



$30 + 55 = 85$

(C)

6. $\frac{d}{dx} \left(\frac{e^x}{e^x + 1} \right)$
 $= \frac{(e^x + 1)e^x - e^x \cdot e^x}{(e^x + 1)^2}$
 $= \frac{e^{2x} + e^x - e^{2x}}{(e^x + 1)^2}$
 $= \frac{e^x}{(e^x + 1)^2}$ (D)

7. $\alpha + \beta = \frac{5}{3}$, $\alpha\beta = -\frac{1}{3}$

From A, $\alpha\beta = -\frac{1}{3}$ NO.

From B, $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$= \frac{25}{9} - \frac{2}{3}$

$= \frac{19}{9} \neq \frac{13}{9}$

From C, $2\alpha + 2\beta = 2(\alpha + \beta)$

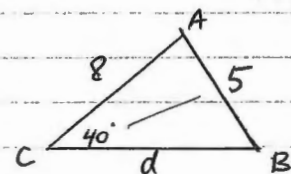
$= 2\left(\frac{5}{3}\right)$

$= \frac{10}{3}$ YES

From D, $\alpha^2\beta^2 = \left(\frac{1}{3}\right)^2 = \frac{1}{9}$

(C)

8.



$5^2 = 8^2 + d^2 - 2 \times 8 \times d \cos 40^\circ$

(D)

9. $r = 2x$ $|k| < 1$

$-1 < 2x < 1$

$-\frac{1}{2} < x < \frac{1}{2}$ (C)

10. 7 (by graphing) (D)

Section II.

Question 11

a) $(2x-7)(x+4)$

2 - correct answer

[1 - signs incorrect eg $(2x+7)(x-4)$]

b) $|x-1| = 2x-3$

$x-1 = 2x-3$

$2 = x$

(Test: OK)

or $-x+1 = 2x-3$

$-2 = 3x$

$x = -\frac{2}{3}$

(Test: fails) RHS = $-\frac{4}{3} - 3 < 0$

$\therefore x = 2$ only

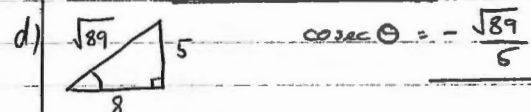
2 - correct solution/testing.

[1 - leaving both]

c) $\int \sqrt{3x-4} \cdot dx$

$= \frac{(3x-4)^{3/2}}{3/2 \times 3} + c$

$= \frac{2}{9} (\sqrt{3x-4})^3 + c$ or $\frac{2}{9} (3x-4)^{3/2} + c$ 2 - correct answer.



$\cos \theta = -\frac{\sqrt{89}}{5}$

2 - correct answer

[1 - finding sqrt(89)]

e) $m_{AB} = m_{BC}$ $\frac{6-12}{8-4} = \frac{k-6}{-4-8}$

$\frac{-6}{4} = \frac{k-6}{-12}$

$k-6 = 18 \therefore k = 24$

2 - correct solution

f) Let roots = $\alpha, -\alpha$ $4(-\alpha) = \frac{4p+1}{2}$

$0 = \frac{4p+1}{2}$

$p = -\frac{1}{4}$

[1 - writing $\frac{4p+1}{2}$ for sum]

2 - correct solution.

g) $f'(x) = 4x - 3 = 5$, at the tangent
 $4x = 8$
 $x = 2, y = 5(2) - 7 = 3$ 1 - finding $x=2$
 $\therefore$ Pt. of contact is $(2, 3)$
 $f(x) = 2x^2 - 3x + c$ (by integration) 1 - finding primitive
 $3 = 2(2)^2 - 3(2) + c$
 $3 = 8 - 6 + c \therefore c = 1$
 $\therefore f(x) = 2x^2 - 3x + 1$ 1 - answer.

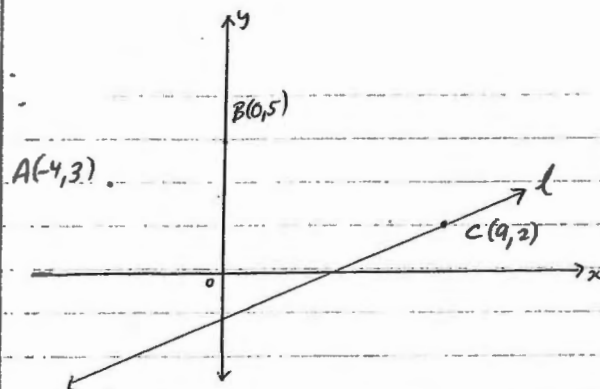
Question 12.

i) $\frac{d}{dx} (x^2 + 4)^5 = 5(x^2 + 4)^4 \cdot 2x$
 $= 10x(x^2 + 4)^4$

ii) $\frac{d}{dx} (e^x \cdot \log_e x) = uv' + vu'$ $\begin{cases} u = e^x & u' = e^x \\ v = \log_e x & v' = \frac{1}{x} \end{cases}$
 $= e^x \cdot \frac{1}{x} + \log_e x \cdot e^x$
 $= e^x \left(\frac{1}{x} + \log_e x \right)$ 2 - correct soln
 [1 - attempt prod. rule.]

iii) $\frac{d}{dx} \left(\frac{\sin x}{x-1} \right) = \frac{vu' - uv'}{v^2}$ $\begin{cases} u = \sin x & u' = \cos x \\ v = x-1 & v' = 1 \end{cases}$
 $= \frac{(x-1) \cdot \cos x - \sin x \cdot 1}{(x-1)^2}$
 $= \frac{x \cos x - \cos x - \sin x}{(x-1)^2}$ 2 - correct soln.
 [1 - attempt quot. rule.]

b)



(i) $AB = \sqrt{(5-3)^2 + (0+4)^2}$
 $= \sqrt{20}$ or $2\sqrt{5}$ units.

(ii) $m_{AB} = \frac{5-3}{0+4} = \frac{1}{2}$

l has $m = \frac{1}{2}$ and through $C(9, 2)$

$y - 2 = \frac{1}{2}(x - 9)$

$2y - 4 = x - 9$

$x - 2y - 5 = 0$

(iii) At D: $y = 0$ $x - 2(0) - 5 = 0$, $x = 5$

$\therefore D(5, 0)$

(iv) $AB \parallel CD$ (given)

$CD = \sqrt{(9-5)^2 + (2-0)^2} = \sqrt{20} = AB$

$\therefore ABCD$ is a parallelogram, since a pair of sides are equal and parallel.

v) $B(0, 5)$
 $l: x - 2y - 5 = 0$

$d = \frac{|Ax_1 + By_1 + c|}{\sqrt{A^2 + B^2}}$

$= \frac{|0 - 2(5) - 5|}{\sqrt{1^2 + (-2)^2}}$

$= \frac{15}{\sqrt{5}} = 3\sqrt{5}$

v.) Area = bh
 $= 2\sqrt{5} \times \frac{15}{\sqrt{5}}$
 $= 30 \text{ units}^2$



The diagram shows a parallelogram with a horizontal base labeled $2\sqrt{5}$. A dashed vertical line from the top vertex to the base represents the height, which is labeled $15/\sqrt{5}$. The area is calculated as 30 units^2 .

Question 13.

a) $\int_1^3 \frac{5x}{x^2+1} dx$
 $= \frac{5}{2} \int_1^3 \frac{2x}{x^2+1} dx$
 $= \frac{5}{2} [\ln(x^2+1)]_1^3$ 1 - correct primitive
 $= \frac{5}{2} (\ln 10 - \ln 2)$ 1 - correct substitution into found primitive.
 or $\frac{5}{2} \ln 5$.

1 - correct substitution
into found primitive.

b) $\Delta = b^2 - 4ac$
 $= k^2 - 4 \cdot 3 \cdot 5 \leftarrow 1$
 $= k^2 - 60 \leq 0 \leftarrow \text{if no real solutions}$

 $-\sqrt{60} < k < \sqrt{60} \leftarrow 1$

c) In $\triangle ADE, \triangle DCG$:

$AD = DC$ (sides of square)

$\angle ADC = \angle GDE$ ($\angle$ s of square)

$\angle ADE = \angle ADC + \angle CDE$

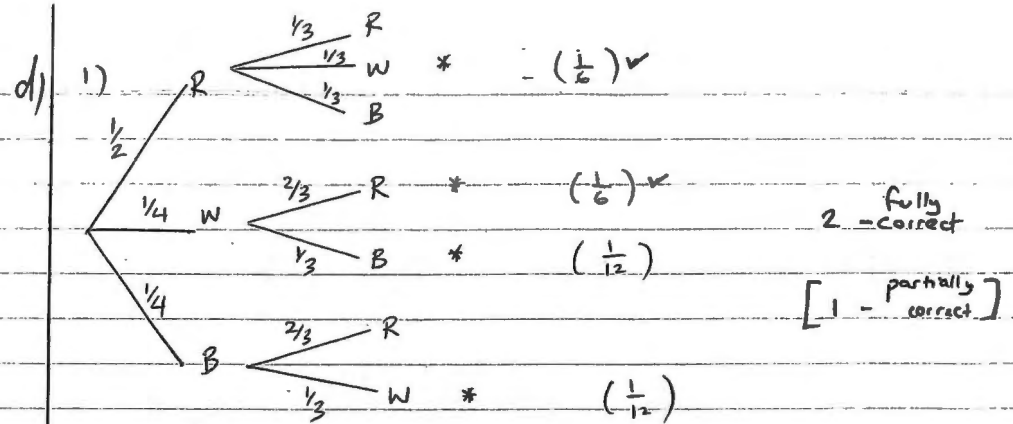
$\angle GDC = \angle GDE + \angle CDE$

$\therefore \angle ADE = \angle GDC$ (adding equal $\angle$ s)

$DE = DG$ (sides of square)

$\therefore \triangle ADE \cong \triangle DCG$ (SAS)

$\therefore AE = CG$ (matching sides of congruent Δ s equal)



ii) $P(RR) = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6} \leftarrow 1$

$$\begin{aligned} \text{ii) } P(\text{red, given white}) &= \frac{1/3}{1/2} \leftarrow \begin{array}{l} \text{outcomes} \\ \text{labelled } \checkmark \end{array} \\ &= \frac{2}{3} \leftarrow 1 \end{aligned}$$

← Outcomes
labelled *

$$= \frac{2}{3} \quad \leftarrow 1$$

$$e) \sum_{n=1}^{20} (3^n + 2n - 1)$$

$$= (3 + 9 + 27 + \dots) + (1 + 3 + 5 + 7 + \dots + 39) \quad \leftarrow \text{split + recognize AP, GP}$$

$$= \frac{a(r^n - 1)}{r - 1} + \frac{n}{2}(a + L)$$

$\nwarrow$ sum of geom. series $\nwarrow$ sum of arith. series

↖ sum of
arith. series

$$= \frac{3(3^{20} - 1)}{3 - 1} + \frac{20}{2}(1 + 39) \quad \begin{array}{l} \leftarrow \text{1 sum of GP} \\ \leftarrow \text{1 sum of AP} \end{array}$$

Question 14

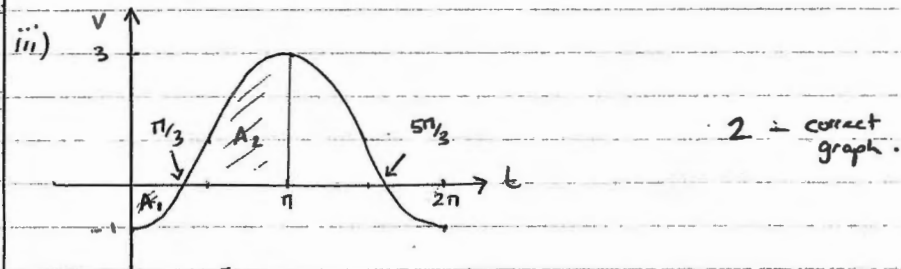
a) i) $\log_3 12.5 = \log_3 \frac{5^{-2}}{2}$
 $= 2\log_3 5 - \log_3 2$

$= 2q - p.$ ← 1

ii) $\log_5 2 = \frac{\log_3 2}{\log_3 5} = \frac{p}{q}$ ← 1

b) i) $v = 0$ when $1 - 2\cos t = 0$
 $\cos t = \frac{1}{2}$ ← 1
 $t = \frac{\pi}{3}, \frac{5\pi}{3} \text{ sec}$ ← 1

ii) $\text{Max } v = 1 - 2(-1) = 3 \text{ m/s}$ ← 1



iv) $A_1 = \left| \int_0^{\pi/3} 1 - 2\cos t \cdot dt \right|$
 $= \left| \left[t - 2\sin t \right]_0^{\pi/3} \right|$
 $= \left| \frac{\pi}{3} - 2\left(\frac{\sqrt{3}}{2}\right) - (0 - 0) \right|$ ← 1

$= \sqrt{3} - \frac{\pi}{3}$

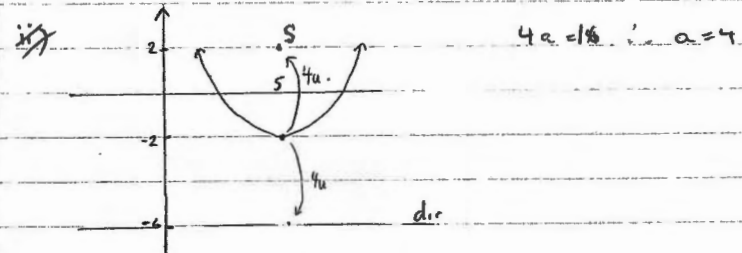
$A_2 = \int_{\pi/3}^{\pi} 1 - 2\cos t \cdot dt$

$= \left[t - 2\sin t \right]_{\pi/3}^{\pi}$

$= \pi - 2(0) - \left(\frac{\pi}{3} - 2\left(\frac{\sqrt{3}}{2}\right) \right)$

$\therefore \text{Distance travelled} = \sqrt{3} - \frac{\pi}{3} + \frac{2\pi}{3} + \sqrt{3}$
 $= 2\sqrt{3} + \frac{\pi}{3} \text{ cm}$ ← 1

c) i) $x^2 - 10x + 25 = 16y + 7 + 25$ ← 1 (completing the sq.)
 $(x-5)^2 = 16(y+2)$
 vertex $(5, -2)$. ← 1



ii) Directrix $y = -6$ ← 1

iii) Focus $(5, 2)$. ← 1

Question 15.

a) i) When $t = 1600$, $m = \frac{1}{2}m_0$

$\frac{1}{2}m_0 = m_0 e^{-1600k}$ ← 1

$-1600k = \ln \frac{1}{2}$

$k = 0.000433$ ← 1 (with correct rounding)

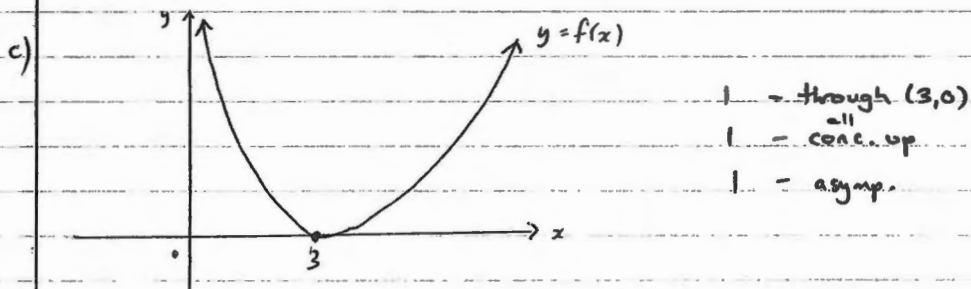
ii) When $t = 1000$, $m = 200$:

$200 = m_0 e^{-1000k}$ ← 1

$200 = m_0 \times 0.6484$

$m_0 = 308 \text{ g}$ ← 1

$$\begin{aligned}
 \text{b) LHS} &= (1 - \cos \theta)(1 + \sec \theta) \\
 &= 1 + \sec \theta - \cos \theta - \cos \theta \sec \theta \\
 &= 1 + \sec \theta - \cos \theta - \cancel{\cos \theta} \cdot \frac{1}{\cancel{\cos \theta}} \\
 &= \sec \theta - \cos \theta \quad \leftarrow 1 \\
 &= \frac{1 - \cos^2 \theta}{\cos \theta} \\
 &= \frac{\sin^2 \theta}{\cos \theta} \\
 &= \frac{\sin \theta}{\cos \theta} \cdot \sin \theta \quad \leftarrow 1 \\
 &= \tan \theta \cdot \sin \theta \\
 &= \text{RHS}
 \end{aligned}$$

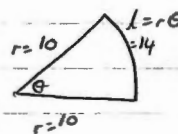


d) i) At A : $x^2 - 5x = x - 5$
 $x^2 - 6x + 5 = 0$
 $(x - 5)(x - 1) = 0$
 ~~$x = 5$~~ or $x = 1$ $\leftarrow 1$
 $\uparrow$
 this is the x -int
 $y = 1 - 5 = -4$
 $\therefore A(1, -4)$

ii) $A = \int_1^5 (x - 5 - (x^2 - 5x)) dx$
 $= \int_1^5 (-x^2 + 6x - 5) dx \quad \leftarrow 1$

$$\begin{aligned}
 &= \left[-\frac{x^3}{3} + 3x^2 - 5x \right]_1^5 \quad \leftarrow 1 \\
 &= \left(-\frac{125}{3} + 75 - 25 \right) - \left(-\frac{1}{3} + 3 - 5 \right) \\
 &= 10\frac{2}{3} \text{ units}^2 \quad \leftarrow 1
 \end{aligned}$$

f)



$$A = \frac{1}{2} r^2 \theta$$

$$= \frac{1}{2} r(l, \theta) \quad \leftarrow 1$$

$$= \frac{1}{2} \times 10 \times 14$$

$$= 70 \text{ cm}^2 \quad \leftarrow 1$$

OR $l = r\theta \therefore 10\theta = 14$
 $\theta = 1.4 \quad \leftarrow 1$

Then $A = \frac{1}{2} r^2 \theta$

$$= \frac{1}{2} \times 10^2 \times 1.4$$

$$= 70 \text{ cm}^2 \quad \leftarrow 1$$

v) If $M=700$ and $A_n=0$:

$$(80\,000 - 200 \times 700) \times 1.005^n + 200 \times 700 = 0 \quad \leftarrow$$

$$1.005^n = \frac{-140\,000}{-60\,000} = \frac{7}{3}$$

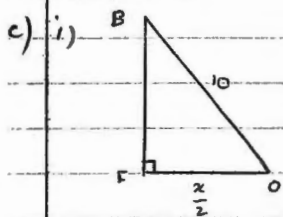
$$n \ln 1.005 = \ln \frac{7}{3}$$

$$n = \frac{\ln \frac{7}{3}}{\ln 1.005} = 170 \text{ months} \quad \leftarrow$$

) $\sin\left(x + \frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$ $\left(0 \leq x \leq 2\pi\right)$
 $\left(\frac{\pi}{6} \leq x + \frac{\pi}{6} \leq \frac{13\pi}{6}\right)$

$$x + \frac{\pi}{6} = \frac{\pi}{3}, \frac{2\pi}{3} \quad \leftarrow$$

$$x = \frac{\pi}{6}, \frac{\pi}{2} \quad \leftarrow$$



$$BF^2 = 10^2 - \left(\frac{x}{2}\right)^2$$

$$= 100 - \frac{x^2}{4}$$

$$= \frac{400 - x^2}{4} \quad \leftarrow$$

$$BF = \sqrt{\frac{400 - x^2}{4}} = \frac{1}{2} \sqrt{400 - x^2}$$

ii) Area = $\frac{1}{2}(a+b)h$

$$\begin{cases} a=x \\ b=20 \\ h=BF \end{cases}$$

$$= \frac{1}{2}(x+20) \cdot \frac{1}{2} \sqrt{400-x^2} \quad \leftarrow$$

$$= \frac{1}{4}(x+20) \sqrt{400-x^2}$$

iii) For max area, $A' = 0$ and $A'' < 0$

$$A' = \frac{1}{4} \left((x+20) \cdot \frac{-x}{\sqrt{400-x^2}} + \sqrt{400-x^2} \cdot 1 \right)$$

= 0 when:

$$\frac{x(x+20)}{\sqrt{400-x^2}} = \sqrt{400-x^2}$$

$$x^2 + 20x = 400 - x^2 \quad \leftarrow$$

$$2x^2 + 20x - 400 = 0$$

$$x^2 + 10x - 200 = 0$$

$$(x+20)(x-10) = 0$$

$$x = -20 \quad \text{or} \quad x = 10 \quad \leftarrow$$

A'' is rather 'messy', so will use A' to test

x	9	10	11
A'	$\frac{1}{4}(324)$	0	$\frac{1}{4}(-37)$
	(+, 0, -)		



$\therefore$ Max area when $x=10$

Question 16

a) i) $A_1 = 80000 (1.005) - M$

$$A_2 = A_1 \times 1.005 - M$$

$$= 80000 (1.005)^2 - M(1.005) - M \quad \leftarrow 1$$

$$= 80000 (1.005)^2 - M(1.005 + 1)$$

$$A_3 = A_2 \times 1.005 - M$$

$$= 80000 (1.005)^3 - M(1.005^2 + 1.005) - M \quad \leftarrow 1$$

$$= 80000 (1.005)^3 - M(1.005^2 + 1.005 + 1)$$

ii) Continuing the pattern

$$A_n = 80000 (1.005)^n - M(1.005^{n-1} + 1.005^{n-2} + \dots + 1.005 + 1)$$

$$= 80000 (1.005)^n - M \times \frac{a(r^n - 1)}{r - 1}$$

where
a = 1
r = 1.005
n terms.

$$= 80000 (1.005)^n - M \times \frac{1(1.005^n - 1)}{0.005} \quad \leftarrow 1$$

$$= 80000 (1.005)^n - M \times 200 (1.005^n - 1)$$

$$= 80000 (1.005)^n - 200M (1.005)^n + 200M \quad \leftarrow 1$$

$$= (80000 - 200M) \times 1.005^n + 200M$$

iii) If $n = 120$ and $A_{120} = 0$:

$$(80000 - 200M) \times 1.005^{120} + 200M = 0$$

$$80000 (1.005)^{120} - (200M) \times 1.005^{120} + 200M = 0$$

$$80000 (1.005)^{120} = 200M (1.005^{120} - 1) \quad \leftarrow 1$$

$$145561.74 = 200M \times 0.8194$$

$$M = \$ 888.16 \quad \leftarrow 1$$